

you're using—Internet Explorer, Mozilla Firefox, Safari, Seamonkey, or anything else—is called a client, which sends a “request” to access the URL you typed in (say www.yahoo.co.in) to your ISP. The ISP forwards the request to another (like a file on the table of a typical government office goes to another table) through the vBNS (the Very high-speed Backbone Network Services) and so on, until it reaches the target where the destination is stored—in this case, the Web site you requested. This “host” computer, or “Web host,” sends back the packet of data you requested through much the same channels in much the same way, hopping routers on the way, until it reaches your ISP. The ISP then sends the data down to your computer and the Yahoo! India homepage appears on your screen.

Simple? No. Actually, mucho, *mucho* complicated.

2.1.6 The Process Explained—Toughly

We'll now try to figure out how the Web works by digging a little deeper and see what exactly happens to get your favourite Web page on your screen.

Let's say you typed in the URL of a Web site. The Uniform Resource Locator, for your convenience, is in letters and numbers, in recognisable words, so that you can identify and recall it at will. The computers aren't human... yet. So, out of our deep sympathy for the dumb machines, we humans have conceded to talk to computers in their language, which is nothing but the language of numbers. Computers converse with each other in the language of zeroes and ones—“bits.” If all computers used this binary language in random ways, exchange of meaningful data would be impossible. To ensure a shared language between two computing machines, we created “protocols.”

2.1.7 Protocols

A protocol is a set of rules that enables the exchange of information between two computers, regardless of whether they run on different operating systems. These rules strictly state the format and procedure to be followed while transmitting data between two computers or two networks. A protocol basically gives the “grammar” to